

Semax Available for Research Use Only

(300-900mcg daily, nasal spray, 2 – 4 week on / 4 – 8 week off)

Semax is used clinically in Russia as an inter-nasal spray for acute ischemic neuroprotection and cognitive recovery, with a good short-term safety record. It shows plausible short-term cognitive benefits, for attention and memory and is generally reported to lack the stimulant-type anxiety seen with classic psychostimulants. It has no long-term safety data and its mechanistic biology suggests aggressive cycling.

 Probably safe (some human evidence)

 Probably effective (strong animal signals and/or some positive human signals)

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

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Semax is a **heptapeptide derived from the ACTH(4–10) fragment (ACTH adrenocorticotrophic hormone is a peptide hormone produced by the anterior pituitary)**, Semax is engineered to preserve the **neurotrophic and melanocortin-linked signaling** of ACTH while eliminating the cortisol-raising endocrine effects of ACTH. Semax is purported to **amplify BDNF-dependent synaptic plasticity, modulate dopaminergic and glutamatergic tone, and enhance cerebral microcirculation**. Semax is intended to function as a targeted neuromodulator that supports cognition, stress resilience, and neuroprotection.

The evidence for Semax's long term human safety is moderate because **small controlled human studies and animal tolerability data suggest short-term safety, but long-term, multi-center safety surveillance is absent** and most trials are regionally concentrated.

The evidence for Semax's efficacy for improving cognitive stress response is good because **mechanistic BDNF/monoamine effects, animal neuroprotection, and small human studies (including fMRI and clinical stroke work) converge on a robust short-term stress-modulating signal**, though larger independent trials are needed.

The evidence for Semax's efficacy for improving calm focus is good because **human pilot data and mechanistic pathways support improved attention and working memory**, but outcome heterogeneity and limited replication reduce certainty.

Dose & Patterns

Intranasal delivery provides rapid CNS uptake. Typical practice-reported regimens are

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- **300 – 900 mcg/day intranasal** (divided doses) with common cycling of 2 – 4 weeks on and 4 – 8 weeks off.

These ranges are **practice-derived from Russian clinical protocols** and community sources and are not universally validated in randomized clinical trials.

Semax (and similar neuroactive peptides) produce benefits by **acutely activating receptors and by engaging slower neurotrophic programs. Cycling is necessary** to preserve those benefits and **reduce adaptive downregulation** of the very pathways the peptide targets. **Continuous receptor stimulation** can trigger rapid cellular adaptations: receptors become less responsive, are internalized, or are downregulated at the transcriptional level. This reduces efficacy over days to weeks for peptides that act via membrane receptors.

Based on Semax's dual actions (rapid monoaminergic modulation plus slower BDNF/NGF-driven plasticity) a **practical, mechanism-informed cycling framework, but not validated in randomized trials might be**

- Increased focus / sustained cognitive enhancement, occasional not sustained use: **1 – (max) 6 weeks / 2 – 4 weeks washout.**
- Maintenance / longevity pattern for long-term use: **repeated 2 – 4 weeks on / 4 – 8 weeks off cycles.**
- For intensive neurorehabilitation (stroke/TBI): **6 – 8 weeks on with 4 – 8 weeks off** and specialist oversight.

However, reports vary widely with some reports using twice daily doses and other reports limiting use to shorter times.

Benefits

- **Improved attention and working memory. Rapid enhancement of attention and short-term memory** reported in **small** human studies and healthy volunteer pilots. *(human clinical data / consistent individual reports)*
- **Neuroprotection after ischemia. Reduced infarct size and improved functional recovery** in **animal stroke models** and clinical stroke series. *(animal testing / human clinical data)*
- **Upregulation of neurotrophic factors. Appears to increase in BDNF and NGF expression** in hippocampus and frontal cortex after dosing. *(animal testing / mechanistic)*
- **Mood and stress resilience. Suggests anxiolytic-like and antidepressant-like effects** in rodent models; anecdotal mood benefits in community reports. *(animal testing / consistent individual reports)*
- **Modulation of monoamine systems — serotonergic and dopaminergic activity changes** that **may support cognition and motivation.** *(animal testing / mechanistic)*

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Indications

- **Acute ischemic stroke recovery and cerebrovascular insufficiency** — used in Russian practice to support rehabilitation
- **Short-term cognitive support (attention, focus) in healthy adults** — commonly used off-label by individuals seeking nootropic effects
- **Adjunctive neurorehabilitation after traumatic brain injury or optic nerve disease** — reported in clinical series

Contraindications

- **Active uncontrolled psychiatric illness (bipolar disorder, psychosis) where monoamine modulation is risky**
- **Pregnancy and breastfeeding** — insufficient safety data
- **Sleep disorders**
- **Concurrent use of MAO inhibitors or unstable cardiovascular disease**
- **Uncontrolled hypertension**

Monitoring

- Blood pressure abnormalities
- Mood during initiation

Stop and seek review for new mania, psychosis, sustained hypertension, or severe insomnia.

Side effects

- **Transient nasal irritation or burning** after intranasal dosing
- **Sleep disturbance or vivid dreams** when dosed late in the day
- **Transient increases in blood pressure (reported with some analogs)** — monitor if hypertensive
- **Mood lability or increased anxiety in a minority with daily prolonged use** (community reports)

Drivers of Diminished Response and Mitigation Strategies

For **Receptor desensitization** the risk is low because Semax is an ACTH fragment that modulates neurotrophic signaling rather than acting as a classical agonist. For **Neutralizing antibodies** the risk is low because Semax is a small peptide with limited immunogenicity reported in the literature. For **downstream pathway adaptation** Semax upregulates brain derived neurotrophic factor and TrkB signaling and preclinical studies show time dependent transcriptional changes that **can reduce responsiveness with continuous dosing**. For **System**

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Level Physiological Adaptation neural plasticity and homeostatic synaptic scaling can **blunt cognitive effects over time. Overall likelihood of waning is moderate.**

Mitigation is **cycling intermittent on off schedules.**

Biological Mechanism

Semax is a heptapeptide derived from the **ACTH(4–10) fragment** that, when delivered intranasally, reaches the brain via **olfactory/trigeminal pathways** and engages multiple neurobiological systems. **Semax upregulates brain-derived neurotrophic factor (BDNF) and nerve growth factor (NGF)** in hippocampus and frontal cortex, supporting **synaptic plasticity, neuronal survival, and memory consolidation**. It also **modulates monoaminergic neurotransmission**, producing **changes in serotonergic and dopaminergic turnover** that can enhance attention and motivation. Additional actions include **interaction with melanocortin receptor signaling** and **inhibition of enkephalin-degrading enzymes**, which together produce **neuroprotective, anti-inflammatory, and stress-buffering effects** observed in ischemia and chronic stress models. These combined trophic and neurotransmitter effects explain the peptide's **rapid cognitive effects** and **slower neurorestorative outcomes** seen with repeated dosing.

Combining Selank and Semax

Combining **Selank** and **Semax** in the same cycle is a common practice in clinics and community protocols because the two peptides have **complementary mechanisms** (Selank: anxiolytic/immune/GABA-monoamine modulation; Semax: neurotrophic/monoamine/melanocortin effects). That complementarity can be useful, but it also raises **additive pharmacodynamic effects** on monoamine systems, sleep, and mood that require careful planning, conservative dosing, and active monitoring. Below I summarize the mechanistic rationale, likely benefits, risks, practical sequencing strategies, monitoring, and example pattern templates — all framed as **practice-derived** options rather than evidence-proven prescriptions.

People combine them because Selank tends to produce rapid anxiolytic and stress-buffering effects via tuftsin-like immune modulation, GABAergic influence, and monoamine balance. While **Semax** emphasizes **BDNF/NGF upregulation** and neuroprotective/memory-enhancing effects while also modulating monoamines. Selank can reduce anxiety and improve stress resilience, which may **unmask or enhance cognitive gains** from Semax (better attention, learning). Semax's trophic effects can consolidate gains made during Selank-mediated windows of lower anxiety and improved focus.

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Potential benefits of combining

- **Faster functional gains with better tolerability.** Selank's anxiolysis can make Semax's cognitive effects more usable in real-world tasks (study, rehab).
- **Broader therapeutic coverage.** You address both **acute neurotransmitter balance** (Selank) and **longer-term neuroplasticity** (Semax).
- **Lower single-agent doses possible.** In some practice patterns, modest doses of each are used together rather than pushing one peptide to a high dose, potentially reducing side-effect risk.

Key risks and interactions to watch when combining

- **Additive monoaminergic effects.** Both influence serotonin/dopamine systems; combined use increases the theoretical risk of **mood destabilization, agitation, or sleep disruption**, especially in people with bipolar disorder or on serotonergic drugs.
- **Sleep disturbance and vivid dreams.** Both can affect sleep architecture; combined late-day dosing increases insomnia risk.
- **Blood pressure effects.** Semax (and some analogs) can transiently affect BP; monitor if cardiovascular risk exists.
- **Unknown long-term safety.** Long-term combined use lacks robust trial data; cumulative exposure should be minimized and periodically reassessed.
- **Formulation and purity variability.** Different suppliers and amidated vs non-amidated variants change potency and stability; mixing formulations increases uncertainty.

Practical combination strategies (mechanism-informed, practice-derived)

1. Sequential blocks (lowest interaction risk)

- **How:** Run a Semax block (focus/neuroplasticity) and follow with a Selank block (mood/stress consolidation), or vice versa.
- **When to use:** If you want to minimize simultaneous monoaminergic load or if psychiatric history is uncertain.
- **Example:** 4–6 weeks Semax on → 2–4 weeks off → 2–4 weeks Selank on → 4–8 weeks off.

2. Concurrent low-dose approach (balanced synergy)

- **How:** Use **lower practice doses** of each peptide simultaneously to gain complementary effects while limiting peak exposure.
- **When to use:** For people with prior tolerance to one peptide, stable mood history, and close monitoring.
- **Practical notes:** Divide daytime doses (morning and early afternoon), avoid evening dosing, and keep total daily exposure conservative.

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3. Alternating-day schedule (reduced overlap)

- **How:** Administer Semax on one day and Selank the next (or Semax mornings / Selank afternoons on alternating days).
- **When to use:** If you want some concurrent benefit but reduce same-day additive effects.
- **Practical notes:** This preserves some continuity while lowering peak combined pharmacodynamic load.

4. Stacked for specific sessions

- **How:** Use Selank on days with high stress or social exposure and Semax on intensive learning/rehab days; avoid using both on the same late-day session.
- **When to use:** For situational optimization (exams, presentations, therapy sessions).

Practical guidance

- **Start conservative.** If combining, begin with **lower doses** and short on-blocks to assess tolerance.
- **Avoid late-day dosing.** Both peptides can affect sleep; schedule doses earlier in the day.
- **Prefer sequencing if psychiatric risk exists.** Sequential blocks reduce simultaneous monoaminergic load.
- **Document effects.** Keep a simple daily log of mood, sleep, BP (if relevant), and cognitive performance to guide adjustments.
- **Always involve a clinician** for anyone with psychiatric history, cardiovascular risk, or concurrent psychoactive medications.

References

Note: Much of the Semax literature is published in Russian journals and conference proceedings; Lebedeva et al. (2018) is a key English-accessible study.

- **Lebedeva IS, Panikratova YR, Sokolov OY, Kupriyanov DA, Rumshiskaya AD, Kost NV, Myasoedov NF.** Effects of Semax on the Default Mode Network of the Brain. *Bull Exp Biol Med.* 2018;165(5):653-656. doi:10.1007/s10517-018-4234-3. (an English-accessible human fMRI study often cited for Semax's rapid CNS effects)
- **Dolotov OV, Karpenko EA, Inozemtseva LS, Seredenina TS, Levitskaya NG, Rozyczka J, Dubynina EV, Novosadova EV, Andreeva LA, Alfeeva LY, Kamensky AA, Grivennikov IA, Myasoedov NF, Engele J.** Semax, an analog of ACTH(4–10) with cognitive effects, regulates BDNF and trkB expression in the rat hippocampus. *Brain Res.* 2006;1116: (short communication). doi:10.1016/j.brainres.2006.07.108. (provides quantitative preclinical data showing Semax-induced increases in hippocampal BDNF and TrkB signaling)

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- Eremin KO, Kudrin VS, Saransaari P, Oja SS, Grivennikov IA, Myasoedov NF, Rayevsky KS. Semax activates dopaminergic and serotonergic brain systems in rodents. *Neurochem Res.* 2005;30(6). (rodent neurochemical study documents acute monoaminergic activation after Semax dosing)